Lecture 11

- Basics of Computer Engineering
- Basics of Electronics Communication

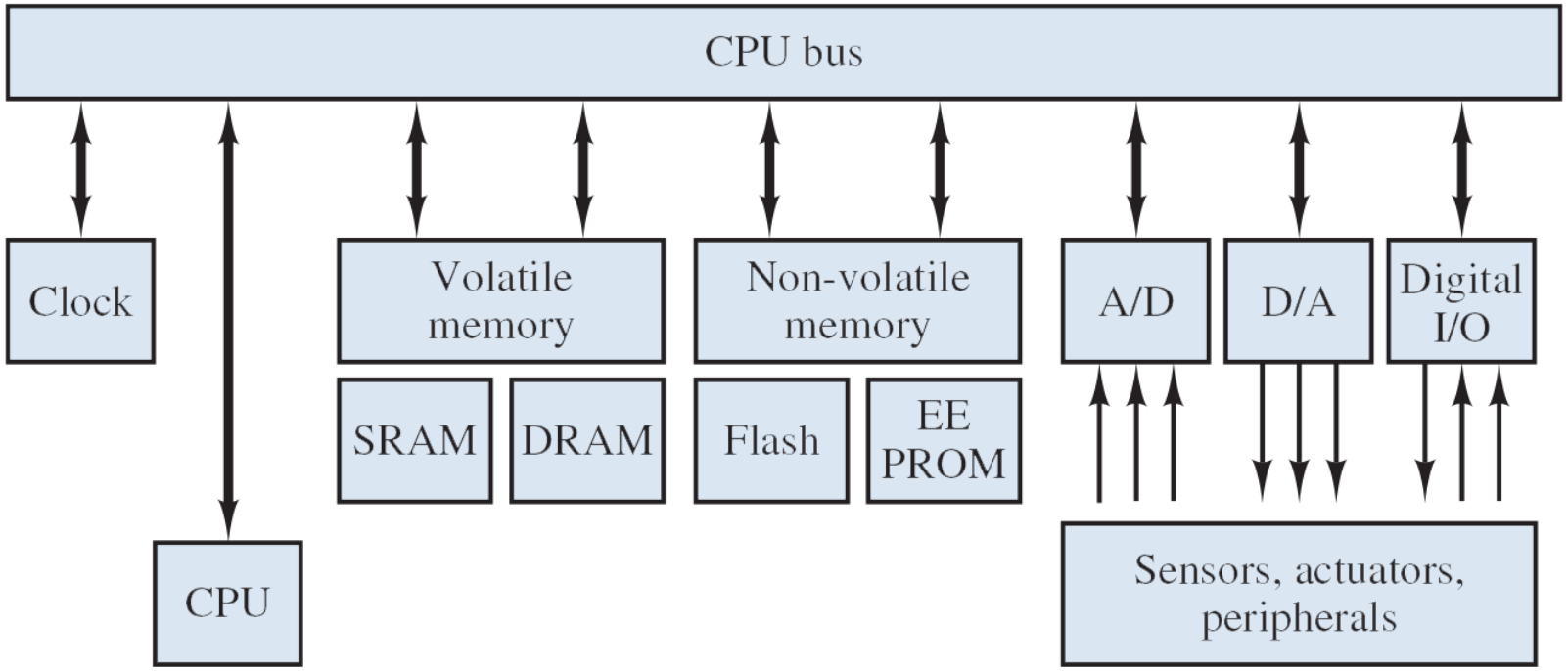
Slide Courtesy: Wikipedia, Google , AI Copilots

How do you send a text message to your friend?

How Your Phone Sends Texts Over the Air ?

Computer System Architecture

- Computer architecture is the design and structure of a computer's fundamental components, including hardware, software, and their interaction.



Central Processing Unit (CPU): The "brain" that executes instructions.

- Performs: **Fetch, decode, execute** cycles.

Memory: Stores data and instructions.

Input/Output (I/O): Interfaces with users/devices (keyboards, displays etc)

Buses: Pathways for data transfer between components.

Core Components of a Computer System

- **Central Processing Unit (CPU):** the "brain" of the computer that executes instructions. **Performs:** Fetch, decode, execute cycles.
- **Arithmetic Logic Unit (ALU):** Handles arithmetic and logic operations.
  - **Control Unit (CU):** Directs operations within the computer.
  - **Registers:** Temporary storage for data during processing.

- **Memory:**

  Volatile memory:

  Primary Memory (RAM): Temporarily holds data and instructions for quick access. SRAM, DRAM

  Cache Memory: Small, fast memory close to the CPU for frequently used data.

  Non-Volatile memory:

  Flash Memory, EEPROM

  Secondary Memory: Permanent storage (e.g., HDD, SSD).

- **Input/Output (I/O) Devices:**
  - **Input:** Keyboard, mouse, sensors.
  - **Output:** Monitor, printer, actuators.

- **System Buses:**
  1. Communication channels for transferring data:
     1. **Data Bus:** Transfers actual data.
     2. **Address Bus:** Transfers location/address information.
     3. **Control Bus:** Transfers control signals.

Clock: a timing device that generates pulses to synchronize and regulate the computer's internal components.

- Clock speed indicates how many instructions the processor can execute per second. (e.g Intel Core i7 Base Frequency 3.80 GHz)

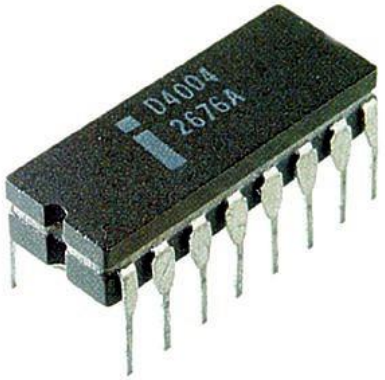
Microprocessors

A **compact CPU** on a single chip that processes instructions for a computer system.

- Combines arithmetic, logic, and control units to execute tasks.

- **Key Features**:
  - **Speed**: Measured in GHz (e.g., 3.2 GHz for fast processing).
  - **Size**: Small enough to fit in phones, watches, or cars.
  - **Examples**: Intel Core i7 (laptops), ARM Cortex (smartphones).

Applications: Laptops, gaming consoles, networking devices, smart appliances.

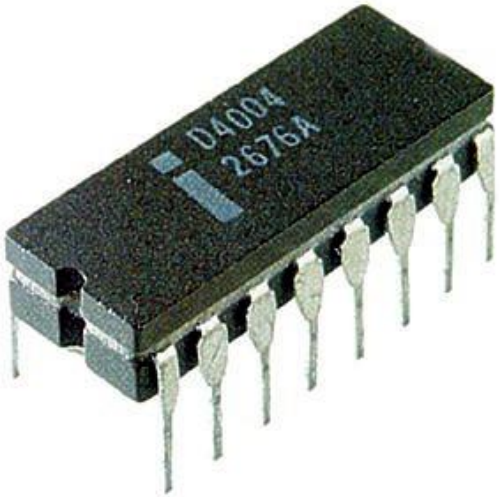


First microprocessor by Intel, the 4004,released in 1971.



Modern Intel Core i7 processor

First microprocessor by Intel, the 4004,released in 1971.



Modern Intel Core i7 processor



Transistors	~2,300	~13 billion (Intel i7-13700K)
Word Size	4 bits	64 bits
Clock Speed	740 kHz	~3.0 – 5.5 GHz (Turbo Boost)

- **From 4-bit to 64-bit**: Modern CPUs handle vastly more data per instruction.
- **From 1 core to many**: Massive parallelism for multitasking and computation.
- **From kilohertz to gigahertz**: ~1 million times faster clock speed.
- **From calculators to AI**: Core i7 can handle gaming, 3D rendering, machine learning, and more.

Embedded Systems

- **What is an Embedded System?**
  - A specialized computer system designed for a specific task, often with a microprocessor.
  - Combines hardware (microcontroller) and software (firmware) to control devices.
- **Key Features**:
  - **Dedicated Function**: E.g., controlling a microwave or car brakes.
  - **Low Power**: Optimized for efficiency (e.g., in IoT devices).
  - **Examples**: Arduino, Raspberry Pi, automotive ECUs.

Applications: Smart thermostats, medical devices, self-driving car sensors.

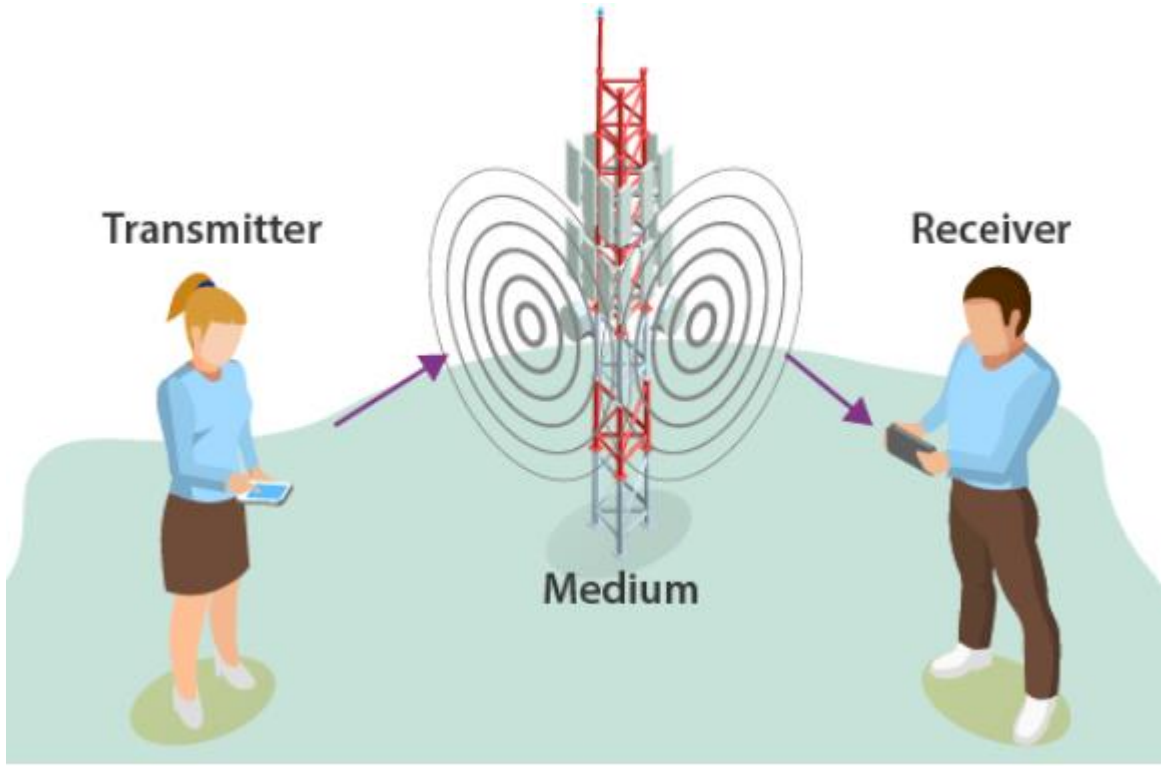
Electronics Communication

- Introduction
- Basic Communication System
- Types of Communication
- Analog and Digital Communication
- Modulation and demodulation
- A brief overview of some modulation techniques

Basics of Communication

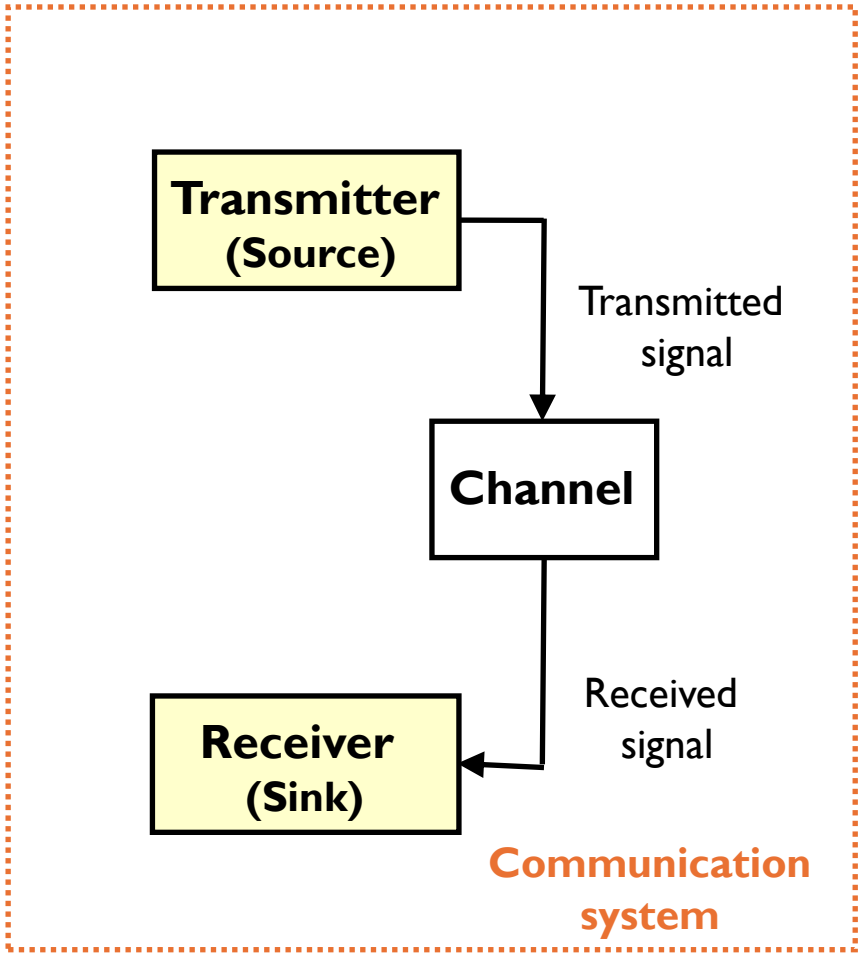
Communication is the transfer of information from one place to another.

This should be done
*- as **efficiently** as possible*
*- with as much **fidelity/reliability** as possible*
*- as **securely** as possible*



Communication System

Communication System: Components/subsystems act together to accomplish information transfer/exchange.



- **Transmitter:** transmits information signal derived from source
  - Signal processing for transmission almost always involves **modulation** and may also include coding.
  - Other functions performed by the transmitter are amplification, filtering and coupling the modulated signal to the channel
- **Channel:** transfers signal from transmitter to receiver
  - Includes analog circuitry
  - Includes propagation medium
  - The channel can have different forms: The atmosphere (or free space), coaxial cable, fiber optic, waveguide, etc.
- **Receiver:** processes received signal to extract the information that was sent
  - Functions include amplification (the received signal may be extremely weak), demodulation and filtering.

Types of Commnication – based on channel

Depending on the communication channel, the communication system is categorized as follows:

1. **Wired** (Line communication)
    - Parallel wire communication (e.g. Early telephone systems)
    - Twisted wire communication (Local Area Networks (LANs) using twisted pair cables (e.g., CAT5 or CAT6 cables for Ethernet).
    - Coaxial cable communication (e.g. Cable TV networks.)
    - Optical fiber communication (e.g. High-speed internet services)
2. **Wireless** (Space communication)
    - Ground wave communication (e.g. AM radio broadcasting)
    - Skywave communication (Shortwave radio broadcasts, HAM radio)
    - Space wave communication (FM radio and television broadcasting, Mobile phone networks (GSM, 4G, 5G) etc)
    - Satellite communication (Direct-to-Home (DTH) television broadcasting (DirecTV), GPS (Global Positioning System) for navigation etc.)

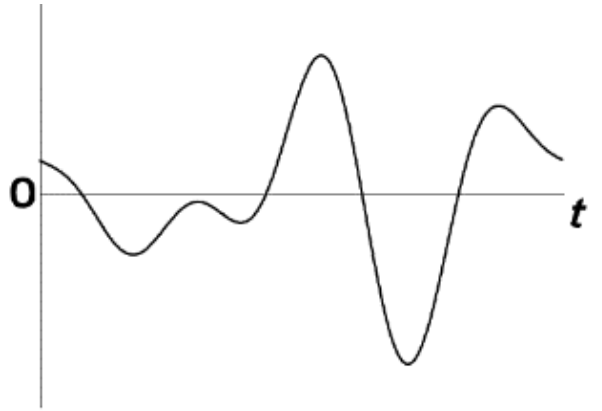
Types of Communication – Analog and Digital Communication

Depending on **signal specification** or technology, the communication system is classified as follows:

Analog Communication:

- Analog communication involves transmitting information using **continuous signals** that vary in amplitude, phase, or frequency over time.
- It is suited for transmitting data in its original, natural form, such as voice or sound.

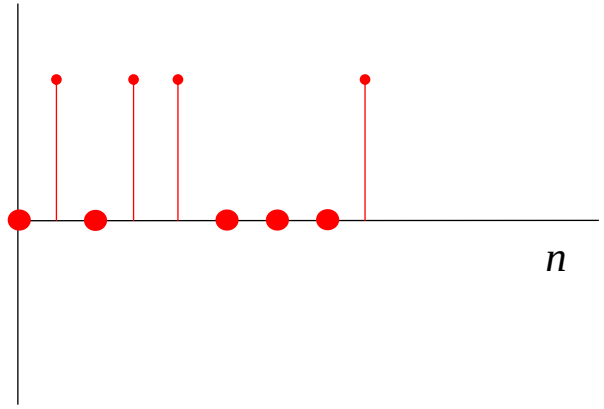
e.g: Traditional AM/FM radio broadcasts, Analog telephones (landlines).



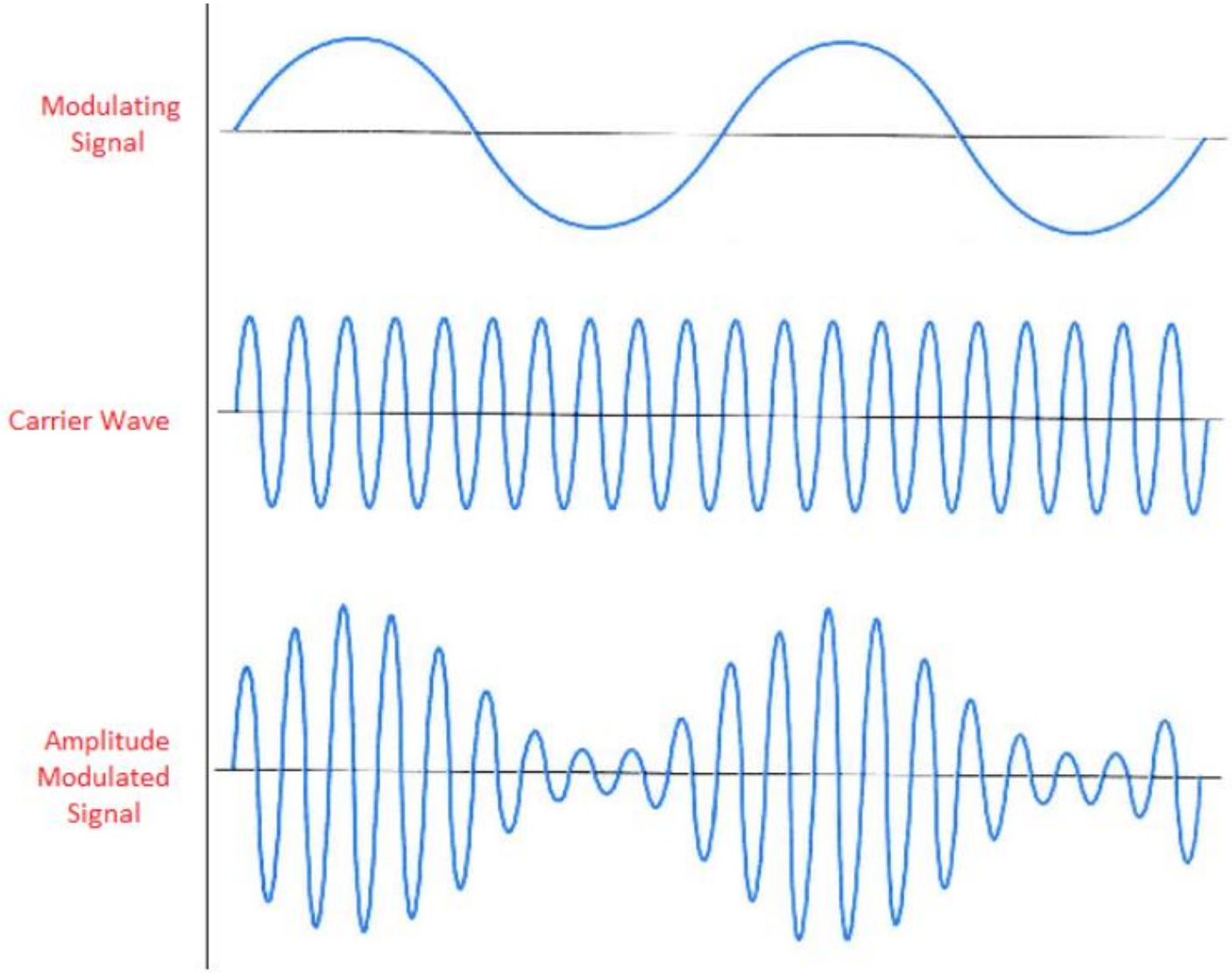
Digital Communication:

- Digital communication involves transmitting information using discrete signals, typically in binary form (0s and 1s).
- It is ideal for encoding, transmitting, and processing data such as text, images, or video.

E.g: Mobile phones using 4G/5G networks, Internet communication (email, video calls).



Modulation and Demodulation



- **Message Signal/ Data Signal/Baseband Signal/Modulating Signal:**

This is the signal that carries the information you want to transmit, such as audio or data.

- The **carrier signal**, typically a high-frequency sine wave, acts as a "carrier" that transports the information to the receiver.
- **Modulated Signal:** The resulting signal after the carrier wave has been altered. This signal contains the information from the modulating signal encoded onto the carrier.

Modulation and Demodulation



Modulation:

- Modulation is the process of modifying a **carrier signal** (a high-frequency signal) to encode the information you want to send, such as voice, data, or video.
- This helps the signal travel long distances over a medium like air, cables, or fiber optics without losing its quality.

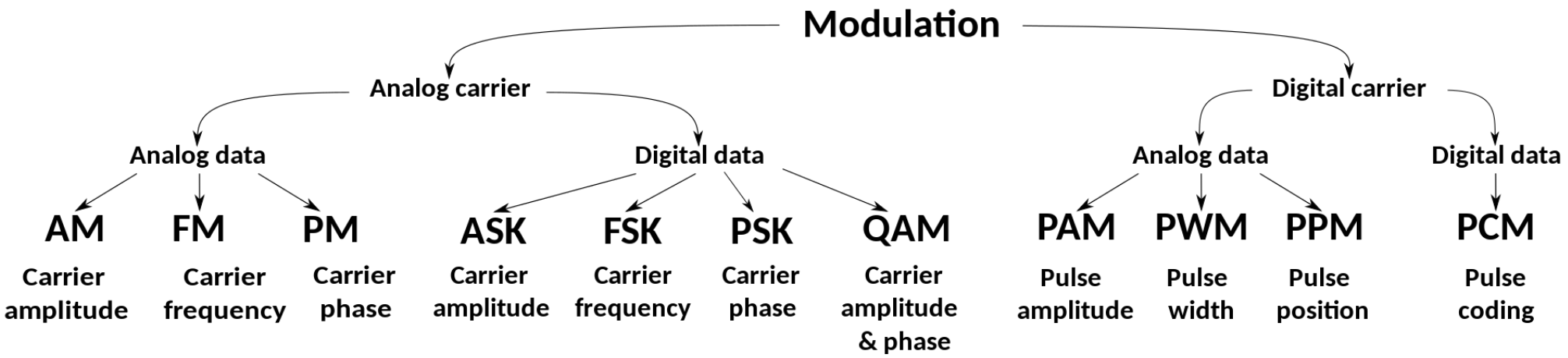
Demodulation:

- Demodulation is the process of extracting the original information from the modulated carrier signal at the receiver's end.
- It reverses the modulation process to retrieve the data sent.

Why Modulation?

- **Efficient Use of Bandwidth**
  - Modulation allows high-frequency signals to carry information efficiently, enabling multiple signals to coexist without interference.
  - Example: FM radio stations operate in different frequency bands (e.g., 88-108 MHz). Modulating the signal onto different carrier frequencies prevents overlap.
- **Signal Transmission over Long Distances**
  - High-frequency carrier waves used in modulation travel further with less energy loss compared to low-frequency baseband signals.
  - Example: AM radio waves use modulation to transmit voice signals over hundreds of kilometers because low-frequency audio signals cannot propagate efficiently.
- **Minimizing Antenna Size**
  - For efficient transmission and reception, the antenna size should be proportional to the wavelength of the signal. Modulating the signal to a higher frequency reduces the required antenna size.
  - Example: A 1 kHz audio signal requires an antenna hundreds of kilometers long, but when modulated onto a 1 MHz carrier wave, a practical-sized antenna (~3 meters) suffices.
  - (Note: For efficient transmission of signals of wavelengths λ the minimum length of antenna should be λ/4. Speed (C)= Frequency (f) x Wavelength (λ))

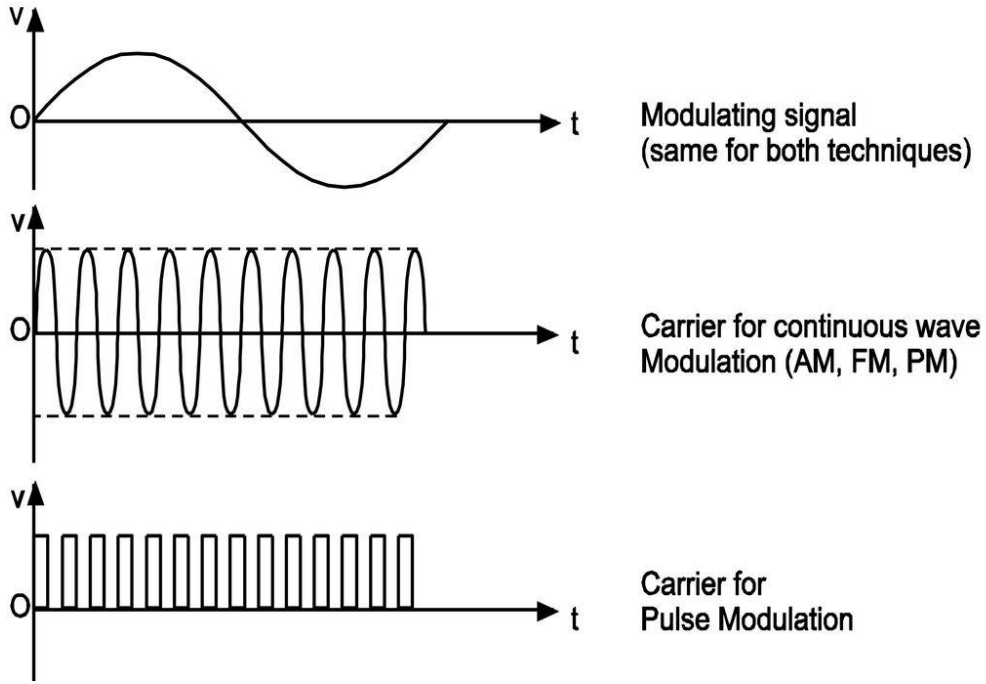
Modulation Techniques



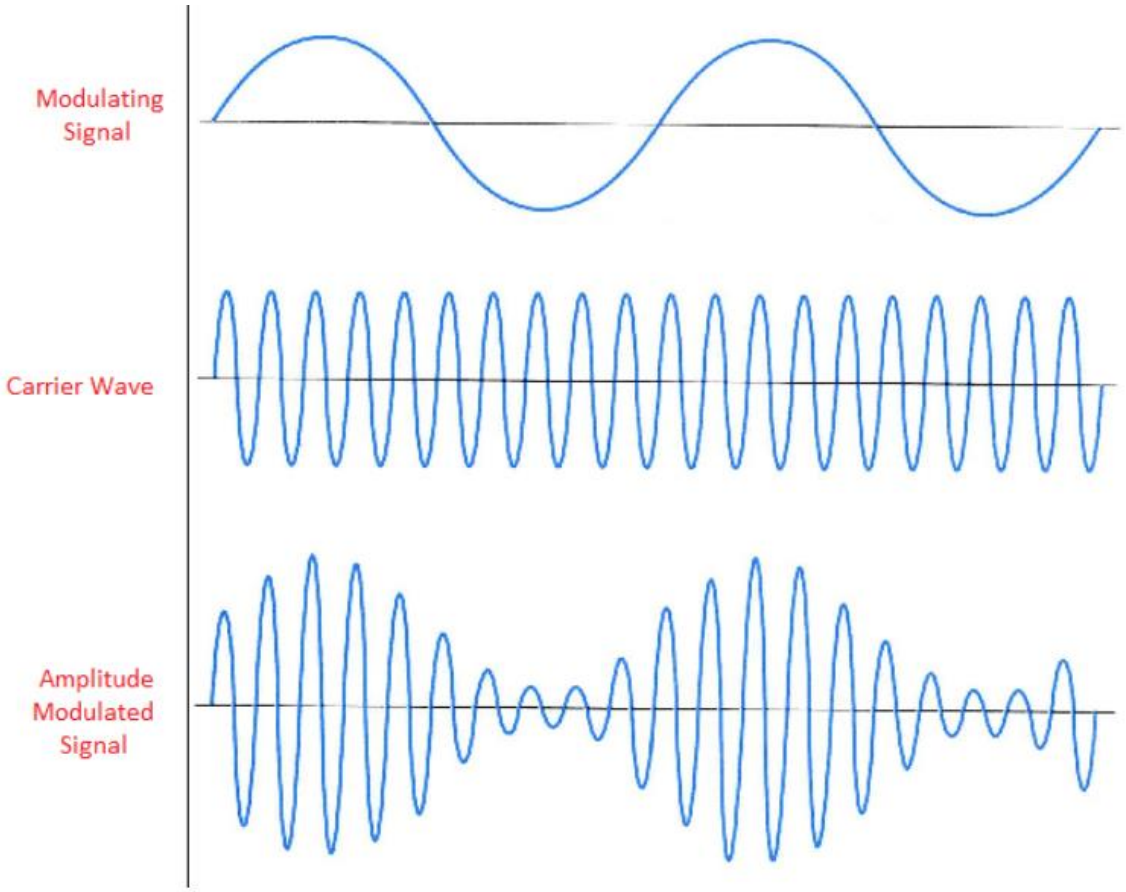
1. Analog Modulation: Data signal is **analog**
   - Carrier is **analog (AM, FM, PM)**
   - Carrier is **pulse (PAM, PWM, PPM)**

2. Digital Modulation: Data signal is **digital**
   - Carrier signal is **analog (ASK, FSK, PSK, QAM)**
   - Carrier signal is **pulse (PCM)**

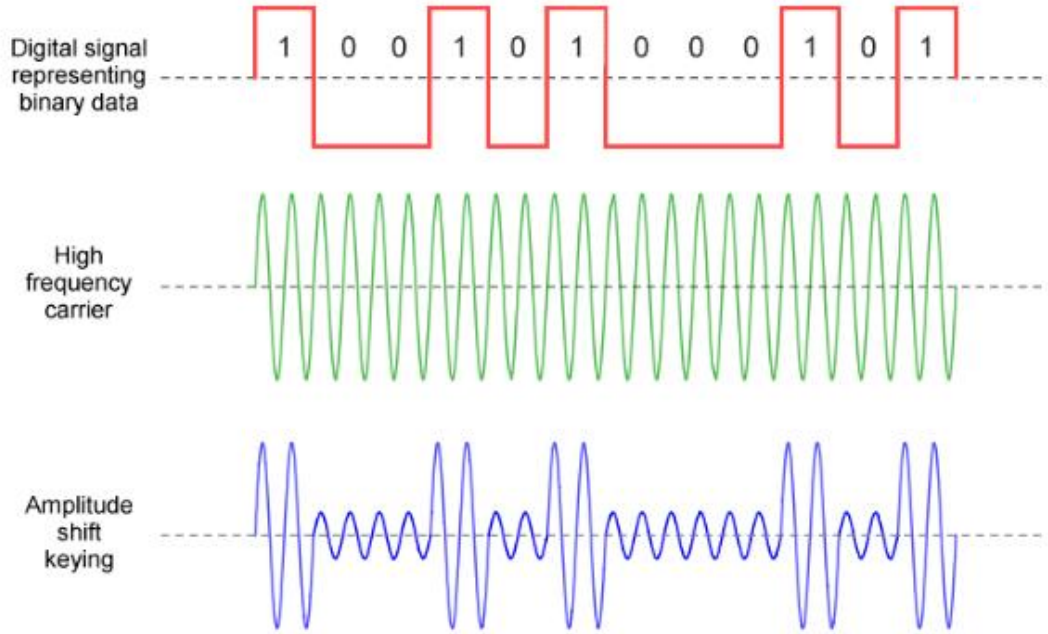
Fig. : Carrier for Continuous Wave and Pulse Modulation



Analog and Digital Modulation examples



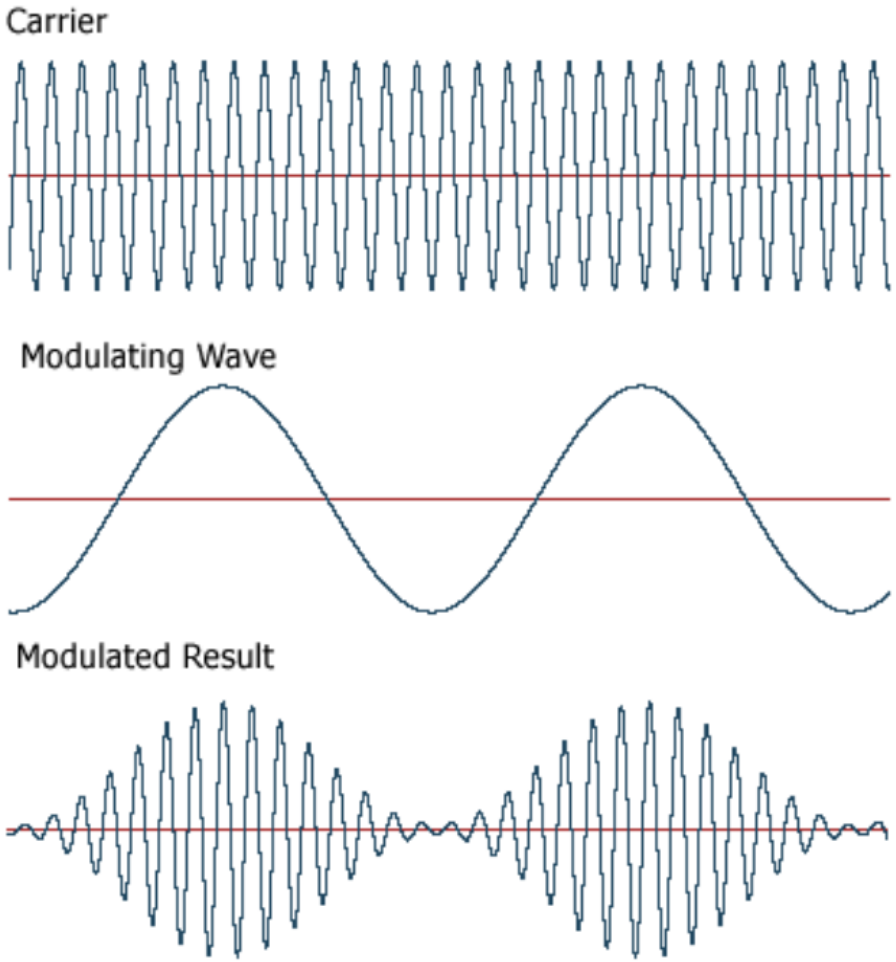
Amplitude modulation (AM)



Amplitude Shift Keying (ASK)

Amplitude modulation (AM)

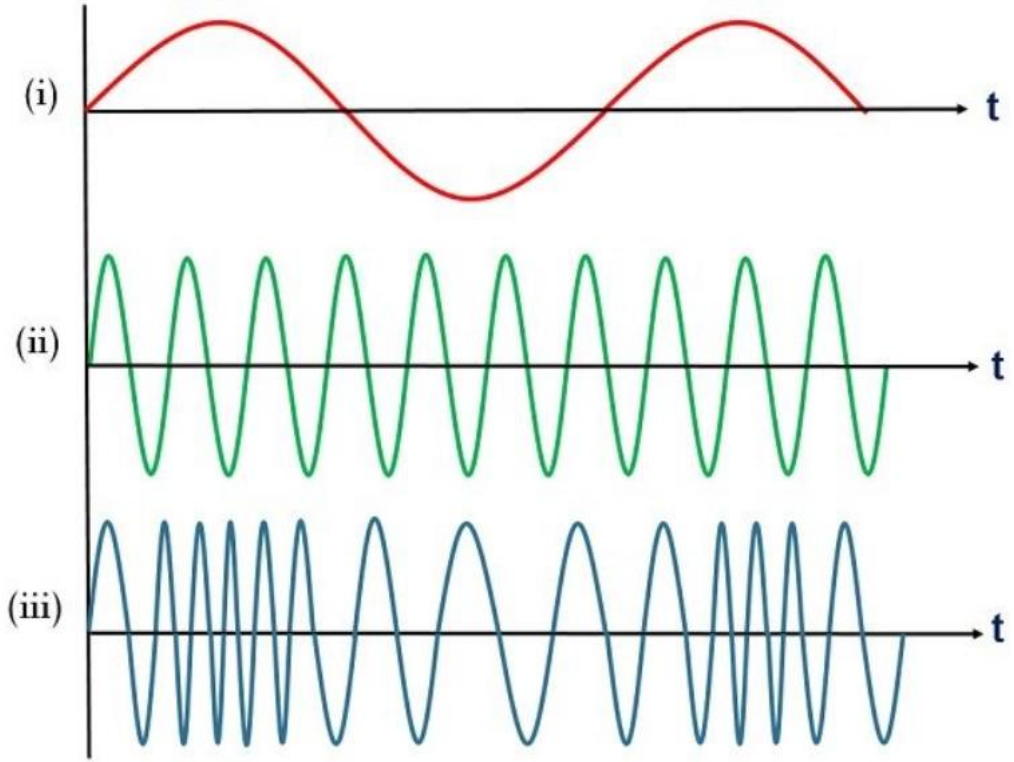
Amplitude modulation (AM):
The amplitude of the carrier wave is varied based on the information signal.
Applications: AM radio broadcasting, aviation communication.
Advantages**:** Simple to implement, lower bandwidth requirements.
Disadvantages: Susceptible to noise, lower sound quality.



Frequency Modulation (FM)

Frequency Modulation (FM)
The frequency of the carrier wave is varied in proportion to the information signal.

- Applications: FM radio, two-way radios.
- Advantages: Better noise immunity, higher sound quality.
- Disadvantages: Requires larger bandwidth than AM.



(i) Modulating signal
(ii) Carrier waveform
(iii) Frequency modulated signal

How do you send a text message to your friend?

How Your Phone Sends Texts Over the Air ?

How Your Phone Sends Texts Over the Air ?



Touch Screen (Embedded System)

- When you type a text message (e.g., “Hey!”), your phone’s **embedded system** converts the text into digital data

Application Processor

- The phone’s operating system packages the text into a format suitable for transmission

Modem: Encoding and Modulation

- The **modem** converts the digital data into an analog signal by modifying a carrier wave

Phone’s Antenna

- The antenna transmits the signal as **electromagnetic waves** (radio waves) to the nearest cell tower.

Cell tower

Receives signal, routes it via core telecom network

Carrier Network



Recipient’s Phone

- The recipient’s phone receives the modulated signal.
- Its **modem demodulates** the signal, converting the radio waves back into digital data.
- The phone’s **microprocessor processes** the data, and the operating system displays the text (“Hey!”) on the screen.

Computer Science Vs Computer Engineering?

- **Computer science** is a more theoretical field that focuses on the study of algorithms, programming languages, and software development. **Focus:** Software, algorithms, and theory.
- **Core Areas of Study**
  - Algorithms and data structures
  - Programming languages (e.g., Python, Java)Software engineering (e.g., app development)
  - Artificial intelligence, machine learning
  - Databases, operating systems, cybersecurity, etc.

Career Paths: Software engineer, data scientist, AI researcher, Web developer, cybersecurity analyst, etc.

- **Computer engineering** is a branch of electrical engineering that focuses on the design and development of computer systems and networks. **Focus:** Designing and developing hardware and software systems.
- **Core Areas of Study**
  - Digital systems and circuit design
  - Microprocessors and embedded systems, Computer architecture (e.g., CPU, memory design)
  - Hardware-software integration
  - Networking and communication systems

Career Paths: Hardware engineer, embedded systems engineer, network architect, etc.